

LAB REPORT

EXPERIMENT 5 : EARTH'S FIELD NUCLEAR MAGNETIC RESONANCE AND MANGETIC RESONANCE IMAGING (EFNMR AND MRI)



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ABSTRACT

Earth's Field Nuclear Magnetic Resonance (EFNMR) and Magnetic Resonance Imaging (MRI) were investigated using the Terranova-MRI system operating at the Earth's magnetic field of approximately 50 μ T. Two samples were studied: a water-filled bottle with two air cavities for system validation and a fresh bell pepper for imaging. Relaxation times were measured using Prospa's native measurement tools, yielding $T_2 = 1480 \pm 100$ ms and $T_1 = 2500 \pm 100$ ms for water, and $T_2 = 590 \pm 10$ ms and $T_1 = 1200 \pm 200$ ms for the bell pepper. The shorter relaxation times observed in the pepper are consistent with the restricted motion of water molecules in biological tissue. 2D spin-echo MRI images of the bell pepper were acquired at resolutions of 16×16 , 32×32 , and 64×64 pixels, demonstrating the fundamental tradeoff between spatial resolution, signal-to-noise ratio, and acquisition time inherent to MRI. The 32×32 image with 5 signal averages provided the clearest visualization of the pepper's internal structure, including its hollow cavity and thick outer walls. These results successfully demonstrate the principles of NMR relaxation and MRI contrast in an Earth's-field system.

INTRODUCTION^[1]

The groundbreaking discovery of X-ray imaging by Wilhelm Röntgen allowed humanity to observe in vivo what was rather impossible before: bones. But with human curiosity, this only brought more questions. What else can we see? Since bones absorb X-rays, how can one see within a skull? MRI takes advantage of the principle that charged particles interact with electromagnetic fields. A proton can only move by either releasing or absorbing photons. And so, if we can force all the protons in one's body to move in a certain way and then measure the

emitted photons, we could reconstruct image slices. Magnetic resonance imaging works this way, with a little more complexity.

When a sample is placed in a strong static magnetic field B_0 , the hydrogen protons within it begin to precess about the direction of B_0 at the Larmor frequency. Because there are many protons and their magnetic moments partially align with B_0 , a net macroscopic magnetization vector $\mathbf{M}$ adds up along the z-axis, parallel to the field.

To extract useful information, a second oscillating magnetic field B_1 is applied perpendicular to B_0 in the form of a radiofrequency (RF) pulse. When this pulse is applied at the Larmor frequency, it tips the net magnetization M away from the z-axis and into the transverse xy-plane. A 90° pulse tips M entirely into the xy-plane, where it precesses and induces a measurable signal in a detection coil. This decaying oscillating signal is called the Free Induction Decay (FID).

Once the RF pulse is removed, the system relaxes back toward equilibrium through two independent processes. The first is longitudinal relaxation, characterized by the time constant T_1 , also called the spin-lattice relaxation time. T_1 describes how quickly the z-component of magnetization M_z recovers toward its equilibrium value as protons release energy to their surroundings. The second process is transverse relaxation, characterized by the spin-spin relaxation time. T_2 describes how quickly the transverse magnetization M_{xy} decays as the protons in the xy-plane lose phase coherence due to interactions with neighbouring spins. These two relaxation processes are independent and generally occur on different timescales, with $T_2 \leq T_1$.

To overcome the rapid dephasing of M_{xy} and extract cleaner signals, a spin echo sequence is used. After the initial 90° pulse tips M into the xy-plane, a 180° refocusing pulse is applied after a time $TE/2$.^[2] This pulse reverses the dephasing of the spins, causing them to rephase and

produce an echo signal at time TE , the echo time. By varying the repetition time TR between pulse sequences, the signal can be made sensitive to either $T1$ or $T2$ contrast, which is the foundation of MRI image weighting.

These principles have had enormous medical impact, allowing clinicians to image soft tissue with high contrast and without ionizing radiation. In this experiment, Earth's Field Nuclear Magnetic Resonance (EFNMR) was used to investigate these principles at a much lower field strength than clinical MRI systems. Rather than the 1.5 to 3 Tesla fields used in hospitals, the Terranova system operates using the Earth's own magnetic field of approximately $50\text{ }\mu\text{T}$ as B_0 . A weird pepper was chosen as the sample due to its high water content and therefore high proton density, making it well suited for producing a measurable NMR signal.

APPARATUS



Figure 1: Photograph of the Terranova EFNMR apparatus with a fresh, weirdly shaped pepper placed inside the transmit/receive coil, ready for NMR signal acquisition. The z-axis is aligned with the yellow arrow, the x-axis is through the cylinder, and the y-axis is perpendicular to both.

The experiment was conducted using the Terranova-MRI Earth's Field NMR system, manufactured by Magritek. This system operates at the Earth's magnetic field of approximately $50 \mu\text{T}$ as B_0 . It consists of a large transmit/receive coil into which the sample is placed, connected to the Terranova control unit. The system uses the Earth's field as the static polarizing field, making it safe and non-invasive. Two samples were used: a water-filled bottle with an air cavity for initial calibration and a fresh bell pepper for the imaging experiment. The bell pepper was chosen for its high water content and therefore high proton density, making it well suited for producing a measurable NMR signal in the Earth's field.

PROCEDURE

Prior to all measurements, electronic devices were removed from the vicinity of the apparatus to minimize electromagnetic interference.

Step 1 — Coil Characterization. Using the *Analyze Coil* function in Prospa, the B1 coil capacitance, inductance, and natural resonant frequency were measured and recorded.

Step 2 — Noise Monitoring. The ambient noise level was monitored in Prospa. Signal acquisition proceeded only after the noise level fell below the 5 μV threshold.

Step 3 — Autotuning. Using the coil parameters from Step 1 as a starting point, the *Autotune* function was used to sweep the capacitance and identify the value that maximized the signal at the Larmor frequency in Earth's field.

Step 4 — Autoshimming. The *Autoshim* function was applied to optimize the homogeneity of the static field B_0 , reducing linewidth broadening of the NMR signal.

Step 5 — Pulse and Collect. A 90° RF pulse was applied, and the free induction decay (FID) was recorded. The linewidth at full width at half maximum (FWHM) was verified to be below 2 Hz before proceeding.

Step 6 — NMR Experiment. With the system tuned and verified, the Earth's Field NMR measurement was carried out.

SOFTWARE PROGRAMMING

All data acquisition and processing was performed using Prospa software (Magritek), which interfaces directly with the Terranova system. Prospa provides built-in pulse sequences, including pulse-and-collect for FID measurement, CPMG for T_2 measurement, inversion recovery for T_1 measurement, and 2D spin-echo for MRI imaging. Data visualization, including signal plots and 2D MRI heat maps, was also performed within the Prospa environment. The **.pt2** files were then parsed in a custom Python GUI [\[3\]](#),[\[4\]](#) to create nicer figures. The **.pt2** binary format stores the 2D magnitude image after a LAER marker as little-endian float32 values. The image is read with **np.frombuffer**, reshaped to $N \times N$, and displayed with **imshow** using **origin="lower"** so the spatial orientation is correct^{[\[5\]](#)}. The function **load_pt2(path)** takes path as an input from a Prospa **.pt2** file and outputs an $N \times N$ NumPy array containing MRI intensity values. See Appendix A for the parsing logic.

DATA AND CALCULATIONS

Table 1

B1 Coil Parameters (Analyse Coil)

<i>Parameter</i>	<i>Symbol</i>	<i>Measured Value</i>	<i>Uncertainty</i>
Capacitance	C	4.1 nF	± 0.02 nF
Inductance	L	0.4044 H	± 0.0004 H
Resonant frequency	F_0	3911 Hz	± 10 Hz

Table 2
System Setup and Signal Quality

<i>Measurement</i>	<i>Value</i>	<i>Uncertainty</i>	<i>Threshold</i>
Noise level	5 μV	$\pm 0.2 \mu\text{V}$	$< 5 \mu\text{V}$
Autotune capacitance	11.5 nF	—	—
Autotune peak frequency	2032 Hz	—	—
FID linewidth (FWHM)	$< 2 \text{ Hz}$	—	$< 2 \text{ Hz}$

Table 3
Autoshim values for the water bottle sample

<i>Parameter</i>	<i>Value</i>
Shim X	1.29
Shim Y	-12.10
Shim Z	-15.33
Larmor frequency (f)	2034 Hz

Table 4**Autoshim values for the bell pepper sample**

<i>Parameter</i>	<i>Value</i>
Shim X	0.90
Shim Y	-13.09
Shim Z	-15.21
Larmor frequency (f)	2034 Hz

Table 5**Measured NMR parameters**

<i>Parameter</i>	<i>Water Bottle</i>	<i>Bell Pepper</i>
Larmor frequency (f)	2034 Hz	2034 Hz
90° B1 pulse duration	0.8868 ms	1.302 ms
180° B1 pulse duration	11.2604 ms	2.387 ms
T2	1480 ± 100 ms	590 ± 10 ms
T1	2500 ± 100 ms	1200 ± 200 ms

RESULTS

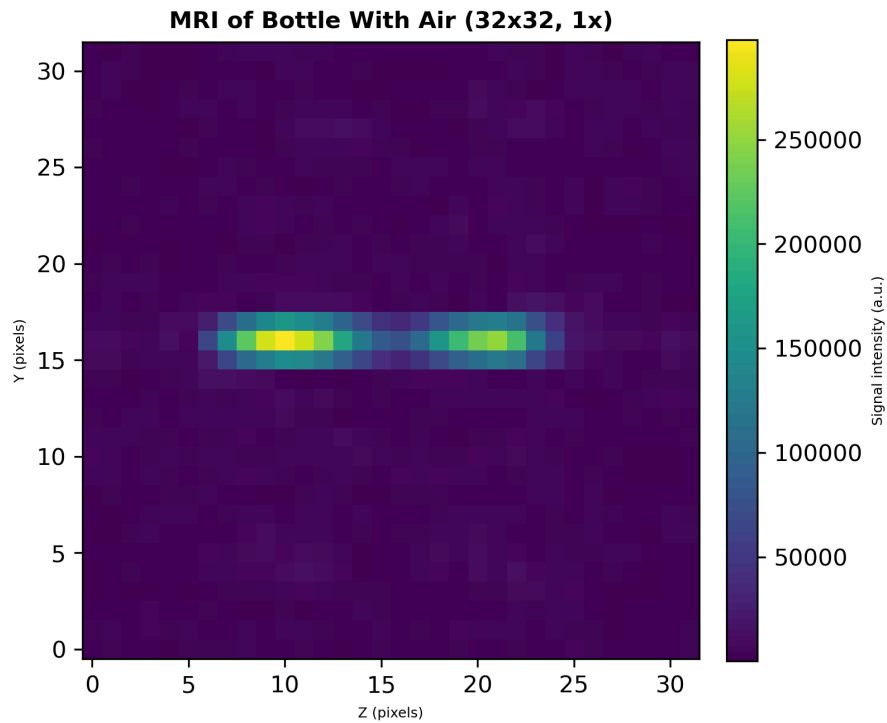


Figure 2: 2D MRI of a water bottle with two air cavities, acquired using a spin-echo sequence at a resolution of 32×32 pixels with 1 acquisition. The two bright horizontal bands correspond to higher proton density regions, while the dark gaps indicate the air cavity, which produces no NMR signal.

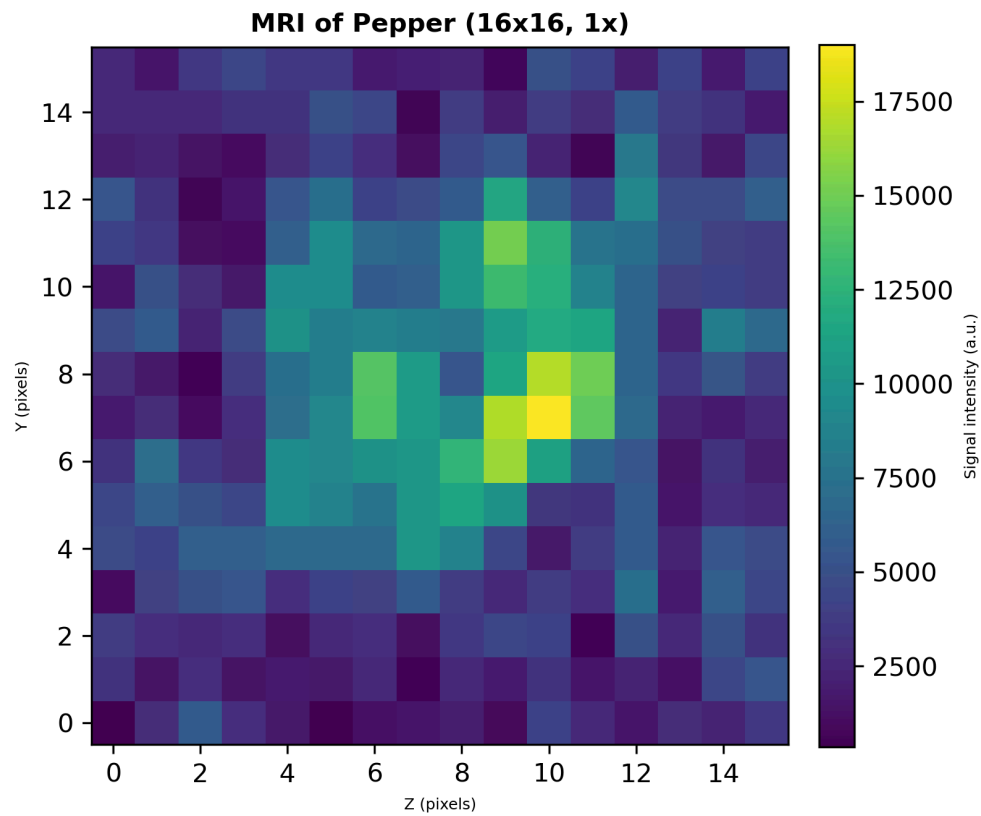


Figure 3: 2D MRI cross-section of the weird pepper acquired at a resolution of 16×16 pixels with 1 acquisition. The image shows a low-resolution representation of the pepper's interior, with brighter regions corresponding to areas of higher proton density.

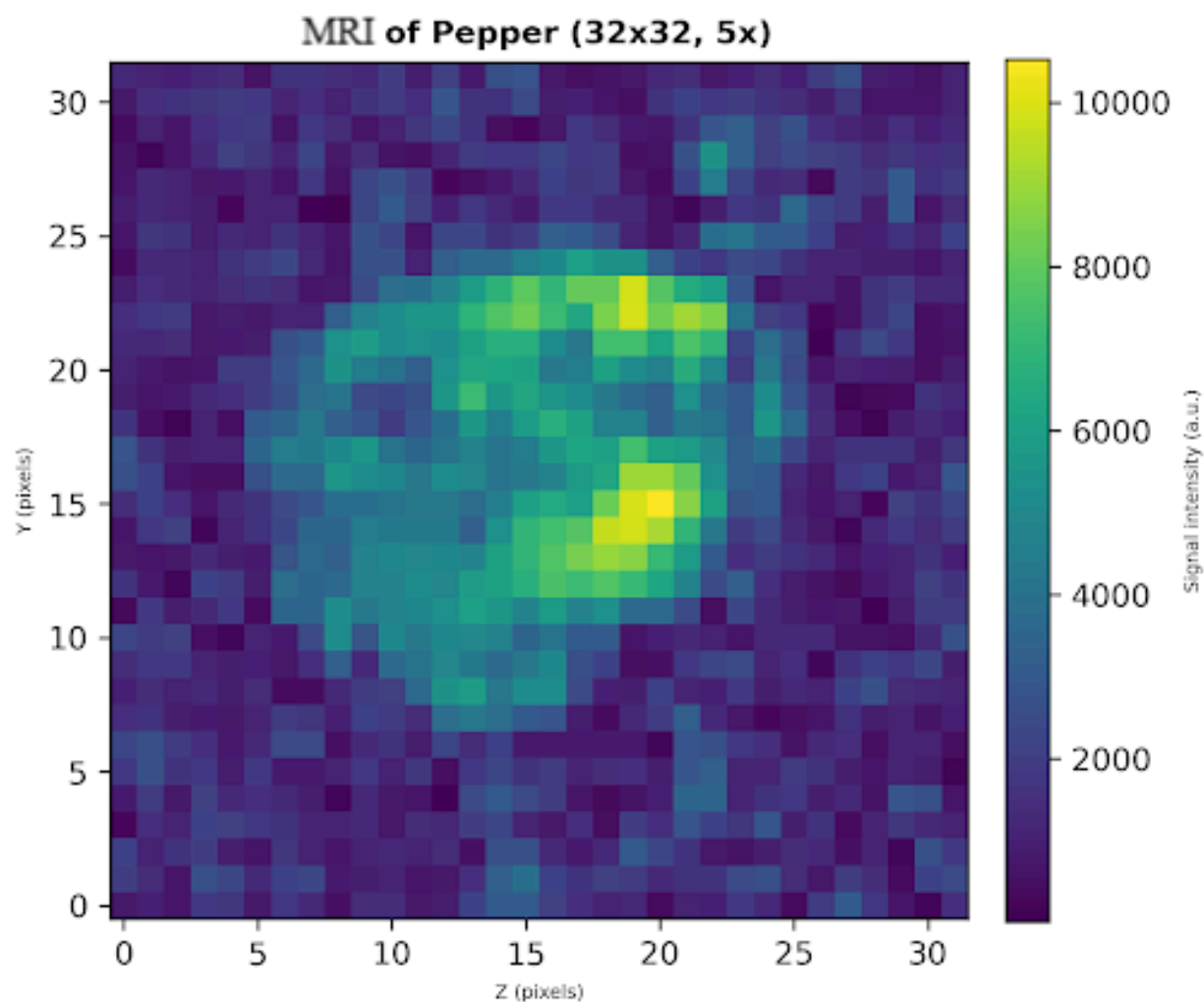


Figure 4: 2D MRI cross-section of the weird pepper acquired at a resolution of 32×32 pixels with 5 signal averages. The increased number of averages improves the signal-to-noise ratio, revealing the hollow interior cavity and thick outer walls of the pepper more clearly. With imagination, the fold of the pepper seen in Figure 1 can also be seen with the bright spots in this image.

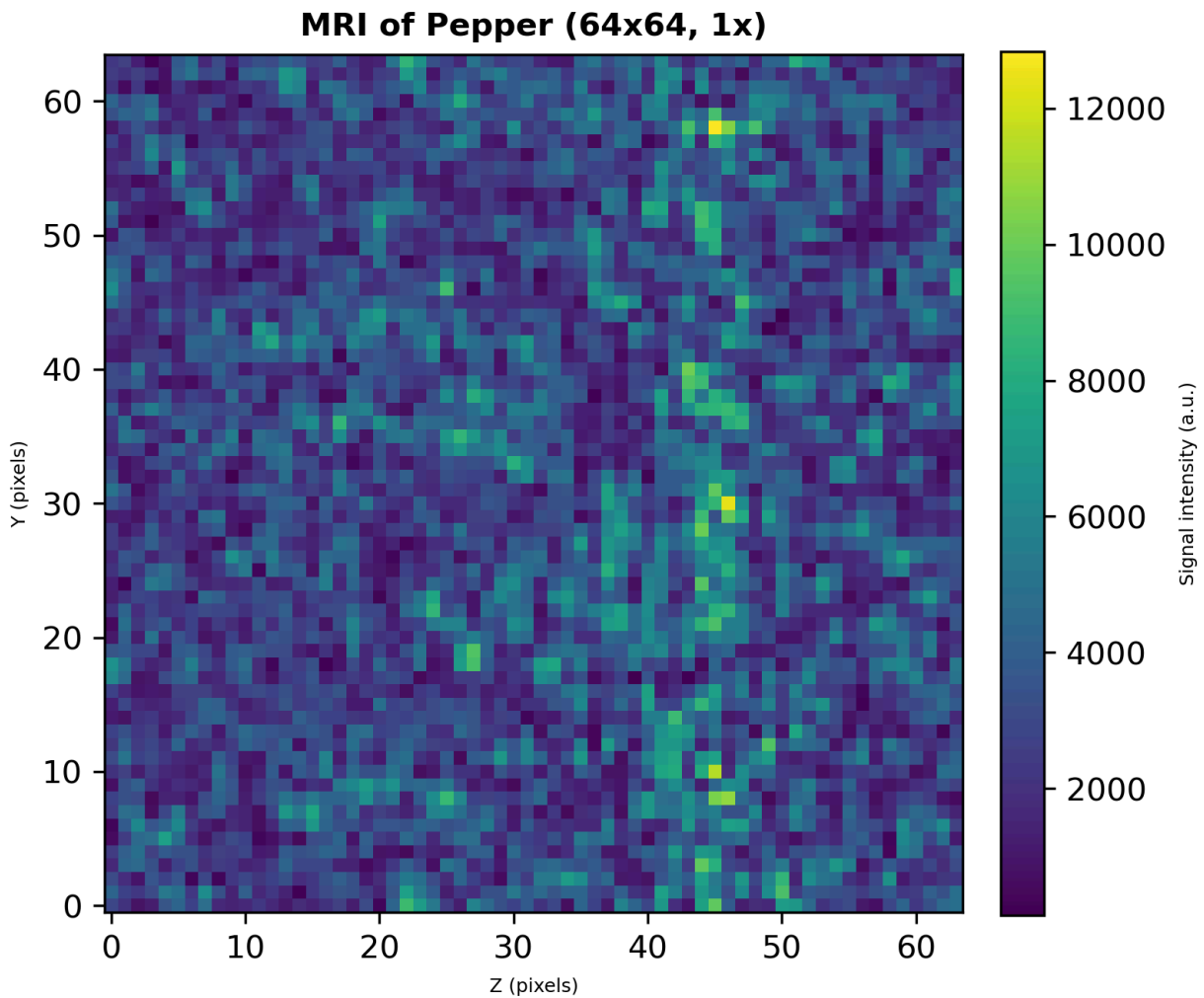


Figure 5: 2D MRI cross-section of a bell pepper acquired at a resolution of 64×64 pixels with 1 acquisition. The higher spatial resolution provides finer structural detail, although the single acquisition results in a considerably lower signal-to-noise ratio compared to Figure 4. As a result, not much can be distinguished from this image.

DISCUSSION

The coil characterization measurements yielded a resonant frequency of 3911 Hz, consistent with the system operating in the Earth's field regime. Following autotuning, the system locked onto a Larmor frequency of 2034 Hz for both samples, confirming that the proton precession frequency was stable across measurements and that the tuning procedure was reproducible.

The measured relaxation times for the water bottle and bell pepper show clear differences consistent with their physical properties. The water bottle yielded $T_1 = 2500 \pm 100$ ms and $T_2 = 1480 \pm 100$ ms, while the bell pepper yielded $T_1 = 1200 \pm 200$ ms and $T_2 = 590 \pm 10$ ms. Both T_1 and T_2 are significantly shorter in the pepper than in free water. This is expected: in biological tissue, water molecules are not free but instead interact with macromolecules such as proteins and cell membranes, which restricts their motion and accelerates both spin-lattice and spin-spin relaxation. The relationship $T_2 \leq T_1$ holds for both samples, consistent with theory.

The MRI images demonstrate the effect of acquisition parameters on image quality. Figure 2 shows the water bottle at 32×32 resolution with 1 acquisition, showing regions of elevated signal intensity consistent with the internal structure of the bottle. This serves as a useful validation of the imaging setup before moving to the more complex pepper sample.

Figures 3, 4, and 5 show the bell pepper at increasing spatial resolutions. The 16×16 image in Figure 3 is heavily pixelated and provides only a rough indication of the pepper's interior structure. The 32×32 image with 5 averages in Figure 4 offers a marked improvement in signal-to-noise ratio, making the hollow interior cavity and thick outer walls of the pepper clearly distinguishable as well as the folded shape of the pepper. The 64×64 image in Figure 5

achieves the highest spatial resolution but at the cost of significantly reduced SNR due to the single acquisition, making it difficult to extract structural information. Together, these three images illustrate the fundamental tradeoff in MRI between spatial resolution, SNR, and acquisition time. Increasing resolution requires either more averages or longer scan times to maintain image quality.

The main sources of error in this experiment include ambient electromagnetic interference, which was mitigated by removing electronic devices and waiting for noise levels to fall below 5 μV before acquisition. Non-homogeneous strength in the Earth's field across the sample volume also contributes to signal dephasing and linewidth broadening, which was partially corrected by autoshimming. At Earth's field strengths, the inherently low polarization of proton spins results in weak signals compared to clinical MRI systems, making the experiment particularly sensitive to noise. These factors together, combined with the image acquisition latency, limit the achievable image resolution and relaxation time precision in EFNMR systems.

REFERENCES

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GitHub. <https://github.com/jjhelmus/nmrglue/issues/146>

APPENDIX A: File Parser For .pt2

```
def load_pt2(path: str) -> np.ndarray:
    """
    Parse a Terranova Prospa .pt2 2D NMR/MRI image file.

    The reconstructed magnitude image is stored after the 4-byte marker
    b'LAER' ('REAL' reversed) as little-endian float32 values.

    Returns
    -----
    np.ndarray
        2D float32 array shaped (n, n).
    """
    with open(path, 'rb') as f:
        raw = f.read()

    pos = raw.find(b'LAER')
    if pos == -1:
        raise ValueError("Not a recognised .pt2 file -- LAER image marker not found.")
    pos += 4
    arr = np.frombuffer(raw[pos:], dtype='<f4').copy()
    n_total = len(arr)
    for n in [16, 32, 64, 128, 256]:
        if n_total == n * n and np.isfinite(arr).all() and arr.max() > 0:
            return arr.reshape(n, n)
    sq = int(np.sqrt(n_total))
    if sq * sq == n_total and np.isfinite(arr).all() and arr.max() > 0:
        return arr.reshape(sq, sq)
    raise ValueError(f"Could not determine image dimensions ({n_total} floats after LAER).")
```